



A Relational Frame Theory-Based Intervention for Improving Reading and Mathematical Competencies Among School Children

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Abstract

The current study examined the effect of an online relational skills training program, strengthening mental abilities with relational training (SMART), on scores on standardized classroom diagnostic assessments. Several previous studies suggested that this novel intervention has a reliable impact on intelligence scores and some domains of cognitive ability relevant in the educational setting. The current study employed a single-blind design and a matched control group in the delivery of the SMART intervention in 45-min sessions 1–3 times per week over approximately 10 weeks, with students completing the intervention within two weeks of each other. Standardized classroom diagnostic assessments were delivered pre- and post-intervention. The results showed that, controlling for baseline scores, the intervention group performed significantly better than the controls in the mathematics and reading domains post-intervention. Gains in both reading and mathematics were large and significant from pre- to post-intervention for the treatment group, but modest significant gains were also found for the control group.

Keywords Relational frame theory · Strengthening mental abilities with relational training (SMART) · Scholastic ability · Curriculum-based measurements · Relational skills · Derived relational responding

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Introduction

Reading and mathematics have been two content areas of great focus for curriculum-based measurements (CBMs) and intervention in K-12 education. In recent years, however, the National Assessment of Education Progress (NAEP) has reported that only 34% of eighth-grade students are proficient in reading, and 34% of eighth-grade students are proficient in mathematics in the USA (National Center for Education Statistics, 2019). NAEP recognizes that the level of proficiency in reading and mathematics has been relatively stagnant for at least a decade (National Center for Education Statistics, 2019). Indeed, there has been limited progress in scores on the National Assessment of Educational Progress in the areas of reading and mathematics since the early 1970s (U.S. Department of Education, 2013), leading to a lack of confidence in the US public school system (PDK/Gallup, 2003a, b, 2016).

This level of academic stagnancy requires the need for more effective intervention strategies (Coddington et al., 2015). One way of proceeding to develop novel and effective interventions to enhance student proficiency in reading and mathematics is to identify specific cognitive skills that may be foundational for these repertoires. To this end, one strategy is to undertake a functional analysis of skill set deficits and to target these systematically using an applied behavior analysis (ABA) treatment approach. Indeed, there is a long history within behavior analysis of attempting to enhance core cognitive skills, particularly language skills (e.g., Hart & Risley, 1968; Risley, 1968; Schreibman & Carr, 1978; Schumaker & Sherman, 1970; Wheeler & Sulzer, 1970; Wolf et al., 1964). That research laid the groundwork for comprehensive intervention programs that in turn evolved into what is now known as “progressive discrete trial teaching.” In the interim, other researchers have drawn more explicitly on Skinner’s analysis of verbal behavior (1957), supplementing a basic behavioral intervention with the specific target of establishing verbal operants as outlined by Skinner (e.g., Sundberg, 2008, 2016; Sundberg & Michael, 2001). Still other researchers have focused on theories of verbal behavior development with a particular emphasis on the generativity of language and the role of speaker–listener interactions, in an effort to accelerate the development of proficient language skills (e.g., Greer & Ross, 2008).

Collectively, early interventions based on the ABA methodology have been found to be promising, although interventions tend to be conducted using cohorts of Autistic children or those with diagnosed learning or behavioral disabilities (see Reichow et al., 2018). Importantly, however, in the context of the current paper, a systematic review by Reichow et al. (2018) found that ABA interventions appear to lead to improvements not only in expressive and receptive speech but on scores on standardized tests of general cognitive ability. As a result, many authors have argued that advances in the training of derived relational responding repertoires might considerably compliment more traditional intervention methods (see Ming et al., 2024). More specifically, researchers making this claim usually draw upon relational frame theory (RFT; Hayes et al., 2001) to develop methods to enhance more complex verbal skills repertoires than are typically targeted (see

Murphy & Barnes-Holmes, 2009; 2010). In order to more fully appreciate these potential advances, we should first provide a precis of RFT.

Relational Frame Theory

The RFT approach to language and cognition starts with the idea that complex human cognitive abilities, such as language and problem solving, are based upon a set of generalized operants known as relational framing skills. More specifically, several core cognitive skills targeted in the educational system rely upon the ability to derive relations between stimuli, or to fluently engage in what we might more colloquially refer to as inferential reasoning. For instance, if a child is shown a cookie and instructed that the word for this object is “cookie” (spoken), and in turn that another word for cookie is “biscuit,” the child will likely be able to respond to the spoken word “biscuit” as equivalent to an actual cookie (e.g., When asked the question, “Show me the biscuit”). This relation is derived in the sense that it was never directly taught by the speaker, but is a relation derived by the combination of two directly taught relations (i.e., cookie—>”cookie” and “cookie”—>”Biscuit”).

While the basic phenomenon of deriving equivalence relations (i.e., syllogistic reasoning) was explicated decades ago by the late Murray Sidman and colleagues (e.g., Sidman, 1971, 2000), RFT takes the additional step of functionally analyzing derived relational responding (DRR) characterized by other types of relationships (e.g., comparison, opposition), and to specify the history necessary for these generalized operants to develop (see Dymond & Roche, 2013). In essence, RFT claims that a range of DRR operants is at the core of human cognition and learning (Hayes et al., 2001; 2009).

From the RFT perspective, a more complete repertoire of derived relational responding than simply deriving equivalence relations includes control over various patterns of DRR by appropriate contextual cues. In other words, a variety of possible relations stimuli (e.g., words and objects) are applicable strictly in appropriate contexts. Another way of putting this is that the relations are arbitrarily applicable depending on context. For instance, if a child is presented with the written word “hot” along with a picture of a fire and another of a snowman and is asked to choose the hot one, they should of course orient toward the picture of the fire and not the picture of the snowman. This establishes that the picture of the fire is in an equivalence relation with the printed word “hot.” However, if the child is asked to point to a picture that is opposite to hot, they should of course orient toward the picture of the snowman and not the picture of the fire. This performance confirms that the choice of picture given the printed word is under appropriate contextual control and that relational responding is flexible and sensitive to contextual cues. Without the establishment of this additional component of the skill, verbal behavior would be stereotypical and insensitive to the contextual nuances of social/verbal interactions.

RFT refers to patterns of relational responding as relational frames, which are pointedly conceived in functional terms as behavioral performances rather than as cognitive constructs. More specifically, behavioral researchers interested in the complex contextual control over relational responding are more interested in the

patterns of responding and the functional properties of the operants being trained than in the topography. For example, a relational frame training intervention would be more concerned with the generalization of a relational operant across numerous novel relata, and then, they would in the establishment of a large number of individually established derived relational responses. In other words, it is the abstraction of relational response, and its unhinging from stimulus topography that renders the response repertoire truly “symbolic” and therefore fit to be used as an explanatory process for a wide variety of acquired skills that can be referred to as intellectual.

Relational responding comes in many forms, such as coordination (Steele & Hayes, 1991), distinction (Roche & Barnes, 1997), opposition (Barnes-Holmes, et al., 2004a, 2004b; Steele & Hayes, 1991), comparison (in terms of any stimulus dimension or property; Dymond & Barnes, 1995), temporality (O’Hora et al., 2008), analogy (i.e., relations between relations; Stewart et al., 2004), hierarchy (Slattery & Stewart, 2014) and deixis (i.e., spatial and/or temporal perspective; McHugh et al., 2004). Combined, the processes of deriving relations and the emergence of contextual control over them produce an outcome whereby an almost infinite variety of relational “frames” can be applied to stimuli under appropriate contextual control. This broader process of arbitrarily applicable relational responding (AARR) is proposed to be a key characteristic of sophisticated intellectual activity. For instance, research focused largely on stimulus equivalence repertoires has strongly suggested the functional necessity of fluent DRR skills for normal language functioning (Barnes et al., 1990; Dugdale & Lowe, 2000), the acquisition of reading skills at a typical rate (de Rose et al., 1992; Farrington-Flint et al., 2007; Mackay, 1985; Sidman, 1971), the emergence of a sufficient vocabulary (Edwards et al., 2011; McHugh et al., 2004; Ming et al., 2024; Nippold & Sullivan, 1987), the development of grammar (Hock, 2003) and spelling (Brown et al., 1996; Mackay, 1985).

Several researchers have considered how a fluent repertoire of AARR might feed upward to enhance complex behavioral skills widely referred to as intelligent behaviors, including scholastic competencies (e.g., McLoughlin et al., 2020a; see also Cassidy et al., 2010). For instance, understanding negative numbers can be conceived in terms of the application of the relation of oppositeness to a positive number but contextually controlled by the negative valence sign. Without a fluent repertoire of responding under the control of oppositeness contextual cues in the vernacular language (e.g., responding to hot as the opposite of cold), it is unlikely that the child will be able to grasp the concept of negative numbers in the context of using mathematical language (i.e., mathematical notation is after all a language). In this sense, several mathematical concepts lie firmly upon what would be widely conceived of as linguistic concepts. At a functional level, however, the skills involved overlap considerably, or may even be synonymous (see Marr, 2015). Indirect support for a functional relationship between AARR and a range of cognitive skills has been suggested by several studies that have reported impressive correlations between various measures of relational framing skills and general or fluid intelligence (e.g., Colbert et al., 2017, 2019; Dixon et al., 2014; Moran et al., 2015).

Importantly, behavioral researchers do not simply assume that AARR is an inherent human ability that in turn explains intellectual and educational attainment, along the lines of Spearman’s (1904) concept of *g*. Rather, RFT aims to

establish the conditions under which this generalized skill itself is established and the degree to which it is malleable. Numerous RFT studies have established that the ability to derive relations is an operant that can be established and enhanced (Barnes-Holmes et al., 2001, 2004a, 2004b, Berens & Hayes, 2007; Dixon et al., 2021a, 2021b; Weil et al., 2011). A solid foundation of such work involving development of methods for establishing relational operants and conceptual work linking AARR skills to a host of cognitive abilities have collectively inspired behavioral researchers to develop means by which enhancements of relational skills might lead to enhancements in general intelligence and scholastic ability. Many of these researchers have explicitly emphasized how these developments build upon rather than replace traditional ABA and Skinnerian methods (e.g., (Dixon, 2015, 2016; Ming et al., 2019, 2022; Sivaraman et al., 2023)). This effort has resulted in a relational account of language and cognition, which has been integrated successfully into several educational and intervention programs (Barron et al., 2019; Cassidy et al., 2011, 2016; Colbert et al., 2018; Hayes & Stewart, 2016; Paliliunas et al., 2022). One such intervention system is PEAK (promoting the emergence of advanced knowledge; Dixon, 2014a, 2014b, 2015, 2016, 2019). The PEAK assessment and training system is a desktop one-to-one based training method that consists of four modules, the first two of which involve direct training and generalization of simple operants relevant to the acquisition of basic verbal skills and the latter two of which involve establishing fluent AARR (in the form of equivalence training and tests for the transformation of response functions in terms of those relations). This system was expressly developed because of the failure of traditional ABA methods to generate more widely generalized change beyond the skills directly trained (see Dixon et al., 2021a, 2021b; May & St Cyr, 2021).

In one recent randomized controlled trial employing 73 Autistic school-age children, Dixon et al. (2023) compared a PEAK intervention focused on DRR with a traditional ABA intervention focused on Skinnerian verbal operants. The PEAK intervention involving relational training (RT) led to greater gains in verbal skills and improvements in standardized intelligence scores (on various tests depending on age) compared to the treatment-as-usual ABA group. The RT group also demonstrated changes in Autism severity indicators, suggesting broader positive effects from the treatment approach (see also Belisle et al., 2018; Dixon et al., 2017; Dixon et al., 2021a, b).

PEAK has been designed for younger children, those with emerging language skills or those with intellectual disabilities. The behavioral approaches, on which PEAK has been based (i.e., ABA), often tailor their programs to work in tandem with supporting challenging behaviors in home or classroom settings, and these have been quite successful. However, another promising relational skills training method called SMART (strengthening mental abilities with relational training; see Guerin et al., 2021; May et al., 2022 for reviews) has been designed specifically for enhancing the cognitive ability of older children who are struggling educationally at school or those that are typically developing. Children who are recommended for SMART are not necessarily Autistic children, who are struggling with their behavior or those in special schools.

The SMART system is based entirely upon the multiple exemplar training (MET) of AARR in accordance with foundational relational frames for childhood education; specifically, coordination, opposition and comparison (see Cassidy et al., 2010). It assumes that the basic repertoires of foundational language have already been established and that there are no severe attentional or other behavioral issues compromising the ability of the individual to undertake the training at regular intervals over an extended period of time (see Roche et al., 2023 for one analysis of boundary conditions in terms of attentional skills; and Wotherspoon et al., 2024 for caveats regarding engagement for specific populations). The system is delivered online across 55 stages of training and testing during which users learn to solve abstract relational puzzles that employ only nonsense words as relata. Figure 1 presents an illustration of a typical training task as presented in the online gamified environment.

Given the behavior-analytic orientation of the intervention, it draws heavily on the use of trial-by-trial reinforcement of correct responses through positive feedback (verbal and auditory) and points (on a fixed-ratio schedule of FR1). It also uses badges as level-by-level rewards to establish engagement. SMART also employs common “gamification” features to enhance engagement, such as time constraints on responding, clear performance targets within and across stages and moderate mental “challenge” (Linehan et al., 2015).

In line with research on the importance of employing ideographically determined levels of game complexity for each user to maximize engagement (Kiili, 2005; see also Davis et al., 2005; Gingold, 2005), the SMART system advances users through the increasingly complex training stages/levels at their own pace, but also accelerates users past testing blocks following training stages that have been mastered without a single error. In addition, by using well-defined complexity parameters (see Method) to dynamically generate tasks at each stage for each user, SMART can



Fig. 1 A sample task within the SMART training protocol. This particular task involves two premise statements which in turn involve different relational frames (i.e., same and opposite). The question probes a derived relation between two relata related linearly from left to right. The yes and no response buttons change location randomly across tasks

deliver true multiple exemplar training to each user and ensure true generalization of those skills to novel stimuli.

The first study using a SMART precursor (Cassidy et al., 2011) employed a small sample of typically developing 10–12 year old children and reported full-scale IQ gains (on the Wechsler Intelligence Scale for Children; WISC-III; Wechsler, 1991) of more than one standard deviation (15 points) on average after several months of training, compared to no gain for a sample of waiting controls. In the second study of that paper, the effect was broadly replicated with a sample of children with a variety of learning difficulties. In a further study, Cassidy et al., (2016; Experiment 1) reported an average IQ gain of around 23 points on WISC-IV full-scale IQ for a sample of fifteen 11–12-year olds, although there was no control group employed. In the second experiment, significant and meaningful gains in scholastic aptitude (as assessed by the Differential Abilities Test; DAT) were reported for a sample of thirty 15–17-year-old adolescents who completed SMART training, which was constituted by gains on both the verbal and numerical subscales.

Hayes & Stewart (2016) replicated this SMART effect using an active control group in a small sample of 11–12-year-old children ($n=28$), reporting significant improvements in standardized assessments of memory, literacy and numeracy following relational training only, and not the control cognitive enhancement method (Scratch coding training). Amd & Roche (2018) administered SMART to a sample of 35 socially disadvantaged children in Bangladesh across several months and were the first to propose a clear “dosage” effect of SMART training with regard to increases in fluid intelligence (assessed using Raven’s Matrices; Raven et al., 2000). A further study involving 15–17-year olds, produced by Colbert et al., (2018), found significant post-SMART IQ gains for all three Wechsler Abbreviated Scale of Intelligence IQ indices (Full-Scale, Verbal and Performance IQ), as well as on each of its four subtests (Vocabulary, Similarities, Block Design & Matrix Reasoning), with no such rises for an IQ-matched control group. More recently, McLoughlin et al. (2020b) conducted a large stratified active-controlled trial to assess the effects of SMART on Non-verbal IQ (NVIQ) in children aged 12–15. A modest, but statistically significant increase in just under 6 NVIQ points was observed for the SMART group only (1.85 points for controls). Similar results were found using a sample of younger children ($n=49$, aged 7–10), with McLoughlin et al. (2020c) reporting an increase in 9 NVIQ points following a SMART intervention in a randomly assigned, actively controlled design study. Importantly, reading test scores obtained at 1-month follow-up for all participants were predicted by post-training measure competent subtasks germane to educational attainment. Importantly, none of the training tasks involved in the intervention were in any way recognizably similar to items on the assessments.

Most recently, a study by Roche et al. (2023) involving a sample of teenage children in Abu Dhabi, broadly replicated previous effects using the brief Wechsler abbreviated scale of intelligence (WASI), but found that for the poorly matched intervention and control groups, controlling for baseline IQ and attentional skills eliminated the observed differences across groups.

A small number of other studies from different laboratories have reported similar positive effects on general cognitive ability as a result of digitized SMART training

(Thirus et al., 2016) or variants on this method (Parra & Ruiz, 2016; Vizcaíno-Torres et al., 2015), albeit using small samples. Still further studies, employing relational skills training to enhance specific cognitive skills, have targeted and reported gains in analogical reasoning (Ruiz & Luciano, 2011), hierarchical reasoning (Mulhern et al., 2017, 2018) and the comprehension of statistics (Sandoz & Hebert, 2016). One recent systematic review has found moderate support for the method in terms of increasing standardized IQ scores, with moderate concerns around study design and experimenter bias (May et al., 2022; see also Beck et al., 2023).

The Current Study

Much of the published literature on the SMART system has primarily focused on intellectual performance and has therefore provided limited data on the program's effect on academic aptitude. The current study sought to assess the utility of SMART as a novel supplemental educational support for improving student attainment in historically low-growth areas: reading and mathematics. It was also important for the current study to employ a pre- and post-intervention measure that was standardized and sanctioned at state level, and widely used for important decision making in a large-scale educational context. For this purpose, the Pennsylvania State Department of Education classroom diagnostic tool (CDT) was employed. The CDT is a "set of online assessments, divided by content area, designed to provide diagnostic information" that is aligned to the Pennsylvania state curriculum standards (PDE, 2015, p. 1). In order to isolate and identify any potential SMART effect, both a treatment group and matched waiting control group were employed in a single-blind study conducted across the intervention period in which both groups continued with traditional educational instruction as normal. It was expected that the SMART group only would experience significant gains in the classroom diagnostic tools (CDT) scores in reading and mathematics post-intervention.

Method

Participants

The study utilized convenience sampling to secure participants from a traditional suburban public high school in southeastern Pennsylvania. All participants were 10th- and 11th-grade high school students, selected based on their enrollment in Algebra and high school English Composition. Seventy-five students were identified on this basis, of which 60 received parental consent to participate. Of these, one failed to attend for follow-up assessments and so was not included in the final analysis. This cohort was divided by a small team of teachers, guidance counselors, and administrators into two matched groups (control, $n=30$, and treatment, $n=29$) based on previous CDT scores. Careful consideration was also given to matching groups in terms of socioeconomic factors, gender and ethnicity as far as possible (see Table 1).

Table 1 Demographic characteristics of participants.

Demographic	Control group		Experimental group	
	<i>N</i>	%	<i>n</i>	%
Gender	30		29	
Male	19	63.3	19	65.5
Female	11	36.7	10	34.5
Ethnicity				
African American	2	6.7	2	6.9
Asian	2	6.7	5	17.2
Hispanic	4	13.3	3	10.3
White	22	73.3	19	65.5
IEP Status				
Gifted IEP (GIEP)	1	3.3	1	3.4
IEP	10	33.3	7	24.1
No IEP	19	63.3	21	72.4
Free lunch eligibility				
No	26	86.7	25	86.2
Yes	4	13.3	4	13.8

Current access to individualized education program (IEP) is indicated along with socioeconomic status, as indicated by entitlement to free school lunches

Materials

Classroom Diagnostic Tools (CDTs)

All participating students took the Pennsylvania's Department of Education's (PDE) classroom diagnostic tools (CDT) assessment at baseline and post-intervention. These assessed both high school literacy and competence in mathematics. The CDT is an assessment of literacy and numeracy that has been reviewed and approved by committees of educators to ensure its alignment with the Pennsylvania state curriculum standards. It provides a snapshot of learning progress throughout the school year and is used by teachers to set individual learning goals, identify deficiencies and assist the reaching of learning targets. This assessment can be administered up to five to six times per year, with a recommended minimum retest interval of five to six weeks. The CDT is an untimed, computer adaptive test that includes 50 to 60 multiple choice questions. A 50- to 90-min time allocation is recommended, with a time-out function in extenuating circumstances. The current assessments were untimed but were completed within 90 min in all cases.

SMART intervention

The SMART system uses a multiple exemplar approach to training, in which users are provided with a very large number of relational problems to solve, using only nonsense words as relata and English words as relational cues (see Cassidy et al., 2016). The basic trial format requires the participant to “solve” logical syllogisms consisting of a varying number of premises specifying relationship between nonsense stimuli, followed by a “yes/no” question pertaining to these premises. For example, a participant may be presented with the relational premise of “FEG is more than BIF,” followed by the question; “Is BIF less than FEG?” Participants may respond to this question by pressing either the “Yes” or “No” button at the bottom of the screen within a 30-s time limit. The nonsense words are always in the form of consonant–vowel–consonant, consonant–vowel–vowel–consonant, consonant–vowel–consonant–consonant or consonant–consonant–vowel–consonant. The nonsense words are chosen dynamically during each task, with the constraint that the same nonsense word is not used twice across any two tasks from the beginning to the end of the entire intervention for a given participant.

The first block (stages 1–29) trains competency in responding to syllogisms involving the concept of sameness/coordination (e.g., “X is same as Y,” “Y is the same as Z,” “Is Z the Same as Y?”) and opposition (e.g., “X is opposite to Y,” “Y is opposite to Z,” “Is Z the Same as Y?”) relations, while the second block targets proficiency in responding to comparison relations (e.g., “X is more than Y” or “Y is more than Z,” “Is Z more than Y?”). Each level begins with a training stage that presents participants with a potentially infinite series of relational tasks and requires the participant to produce 16 consecutive correct responses in order to progress to the testing stage. During this training stage, corrective feedback for all responses is provided (i.e., the words “Correct” or “Wrong” appear on screen for 1 s accompanied by a beep or a bloop sound, respectively). Upon reaching the fluency criterion, participants continue to a testing stage in which a series of 16 novel relational tasks of the same level of complexity as used in the preceding training are presented, but without corrective feedback. All tests require the participant to respond correctly to each of the 16 tasks presented, in order to move on to the next level of the program. If the participant produces an incorrect response on any of the test trials, they are recycled back to the training stage to repeat this process (with novel task items) until the passing criterion is satisfied once more, after which they are again presented with a test (always involving yet more novel task items and stimuli). This recycling process may continue across a potentially infinite number of training and testing trials, until they satisfy the progression criteria necessary to move on to the next level and, eventually, complete the program’s 55 levels.

There is a 30 s time limit on responding for each task. If a participant fails to respond within this period, this is treated as an incorrect trial and the screen will briefly display the message: “You must respond more quickly.” During each trial, a countdown timer is clearly visible on the screen above the relational task window. During the last 10 s of the 30 s response window, the normally white countdown timer changes to red.

Cassidy et al (2016) outlined that the increasing complexity of the task types across the 55 stages is determined by;

- The number of premise statements presented to the participant before the relational question (one, two or three).
- The order of the premises
- The nodal distance between the relata. For example, given the premises, A is the same as B, B is the same as C, and C is the same as D, the question; Is B the same as A? represents a zero-node derived relation probe. The question, Is C the same as A?, represents a one-node derived relation probe, and the question, Is D the same as A?, represents a two-node derived relations probe. Generally, the greater the nodal distance between the two relata referred to in the relational question, the more complex the relational task.
- The directionality of the relational question. For instance, given that A is more than B, B is more than C, and C is more than D, correctly answering the question; Is D more than A? (i.e., reverse direction to that of the linear logic of the premise sequence), involves a more highly developed relational skills repertoire than correctly answering the question, Is A more than D?
- Whether or not more than one relation type was specified across the premises, and
- Whether or not the relational term employed in the question was employed in any of the premises.

Relational Abilities Index

A relational abilities index (RAI) test was administered as part of the SMART training both before and after the intervention. The RAI (Colbert et al., 2017) was designed as a measure of relational framing proficiency that maps directly on to the 55 training stages delivered in SMART (i.e., each of the SMART's 55 levels is represented by a single assessment trial in the RAI). The test is simply scored as the total of correct responses out of 55, and this score is not transformed. Thus, the RAI can function as a manipulation check for SMART training insofar as effective training should lead to increased scores on the RAI post-intervention. The stimulus generation procedure outlined above for SMART is replicated within the RAI, and thus, no two RAI administrations are formally identical, eliminating the possibility of practice effects.

Procedure

Participants had been assessed by teachers using the CDTs earlier in the academic year as part of standard classroom practice. This baseline assessment was therefore conducted blind to the study parameters. Teachers, guidance counselors and administrators assigned students to the treatment or waiting control group based on these scores, while controlling for socioeconomic factors, gender and ethnicity, as

far as possible. Teachers then supervised the administration of training in 1–3 sessions per week (approx. 30–45 min) in a group setting. Students were encouraged to continue training at home between sessions as part of their homework, in order to reach around three hours of training per week, or to complete the training within 12 weeks. However, sufficient progress was made in the supervised sessions that very little training occurred outside the school setting. Following the 12 weeks of training (excluding a break period of two weeks), the post-intervention CDTs were again delivered by teachers. The CDT assessment and scoring was automated. However, various teachers transcribed the results of different students, linked only to anonymized student identification numbers.

When all students in the treatment group had completed training (within a couple of weeks of each other), the waiting control group were provided with access to the SMART software for 12 further weeks, but CDT assessment was not administered post-training (Fig. 2).

The SMART program began with a RAI pre-test that provided a baseline measure of relational skill fluency, and this was again automatically administered within the training software after the completion of stage 55. Student online training progress was monitored by the principal researcher using a teacher control panel provided by the software developers. Feedback was provided to the host school regarding student progress, and this served to maintain adherence to the training program.

Results

Mean pre- and post-intervention RAI scores as well as change scores can be seen in Table 2.

Treatment Group



Matched Waiting Controls

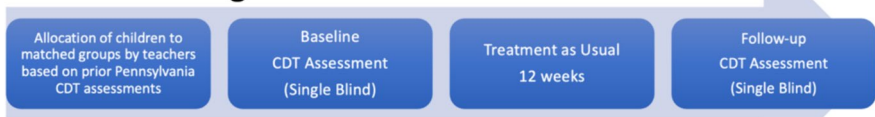


Fig. 2 Chart illustrating the procedural sequence for both the treatment and control groups

Table 2 Relational abilities index scores out of a maximum score of 55 (treatment group only)

	<i>N</i>	Mean	Min	Max	Standard deviation	Variance	Median	Mode
Experimental group								
RAI pre-intervention	29	43.31	24	55	8.14	66.28	45.0	48
RAI post-intervention	29	50.55	37	55	4.52	20.45	52.0	55
RAI change	29	7.24	0	20	5.68	32.25	6.0	1

Table 2 shows an average gain of 7.24 RAI points, which is comparable to gains seen in SMART intervention studies to date and in the region of twice the test–retest differential (Colbert et al., 2017). Increases in scores were observed for each of the participants, and this increase was significant at the group level ($z=4.46$, $p<0.001$), indicating the successful manipulation of the independent variable (i.e., relational skill proficiency).

All of the participants that began the SMART program completed the program in its entirety (55 stages) within 45 school days (approx. 10 weeks), 15 school days short of the planned 60-school day study period. This indicates exceptional adherence to the training protocol. The scores for both the pre- and the post-intervention CDT assessments were scaled out of 100 percent and are summarized in Table 3. Table 3 outlines the mean pre- and post-intervention scaled CDT scores separated out for mathematics and reading aptitude for both groups, as well as gain scores in each case. Mean gains in both mathematics and reading scores were considerably

Table 3 Mean mathematics and reading scores across time for both groups, along with measures of central tendency

	<i>N</i>	Mean	Min	Max	Standard deviation	Variance	Median	Mode
Control group								
Pre-test mathematics score	30	61.20	30	94	16.22	263.3	59.0	52
Pre-test reading score	30	55.60	37	83	11.51	132.5	52.5	50
Post-test mathematics score	30	63.50	40	95	14.30	204.6	62.0	62
Post-test reading score	30	57.33	37	83	11.66	136.0	54.0	63
Mathematics growth	30	2.30	−6	10	4.50	20.2	2.0	2
Reading gain	30	1.77	−6	13	4.36	19.0	2.5	3
Experimental group								
Pre-test mathematics score	29	64.83	28	98	18.83	354.4	66.0	50
Pre-test reading score	29	52.72	20	93	19.76	390.4	50.0	43
Post-test mathematics score	29	73.93	34	100	16.66	277.6	76.0	66*
Post-test reading score	29	64.03	33	93	16.26	264.2	65.0	53
Mathematics gain	29	9.24	−6	30	9.33	87.0	6.0	6
Reading gain	29	11.24	−10	36	12.19	148.6	12.0	0*

* multiple modes exist, smallest value is shown

higher for the treatment group (9.24 and 11.24, respectively) compared to the controls (2.30 and 1.77, respectively).

A between-groups analysis of variance (ANOVA) comparing the treatment and control groups found them to not significantly differ from each other in baseline mathematics scores ($[F(1, 58) = 0.63, p = 0.431]$) and baseline reading test scores ($[F(1, 58) = 0.471, p = 0.496]$). Welch ($p = 0.432$) and Brown–Forsythe ($p = 0.432$) post hoc tests confirmed that this effect maintained under nonparametric assumptions.

A one-way analysis of covariance (ANCOVA) was used to determine group differences in post-intervention CDT scores, while controlling for pre-intervention scores. This analysis was run with reservations given a nonparametric distribution of the pre-intervention CDT scores for reading. Performance in the mathematics post-intervention CDT was significantly higher for the treatment group, $F(1, 58) = 19.68, p < 0.001$. The effect size was calculated using Cohen's d as $d = 1.19$ (large effect). A similar and significant effect was observed for the post-intervention CDT reading scores across groups, $F(1, 58) = 17.25, p < 0.001$, where Cohen's d was found to be 1.11 (i.e., a large effect; see Fig. 3).

Planned comparisons were run to check if the gains were significant across time within each group. Paired sample t tests were used in all cases, and a Bonferroni-corrected alpha was set to 0.125. Interestingly, significant improvements in the mathematics content area were observed for both the treatment, $t(57) = 28.6, p < 0.0001$, and control groups, $t(59) = 31, p < 0.0001$, although a medium effect size was observed for the treatment group ($d = 0.5$), compared to a small effect size observed for the control group ($d = 0.15$). Similarly, significant improvements in the reading content area were observed for both the treatment, $t(57) = 23.2, p < 0.0001$,

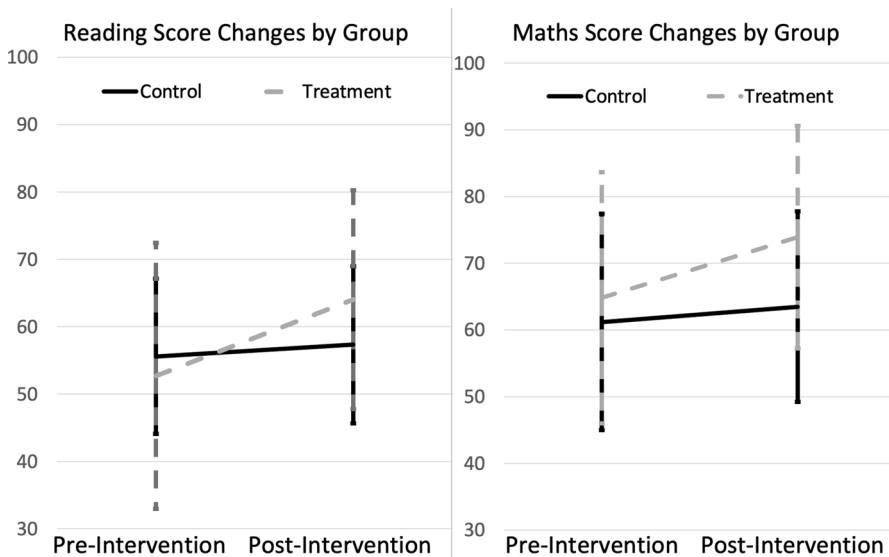


Fig. 3 Reading and mathematics scores changes from pre to post-intervention for both groups. Vertical bars represent standard deviations displayed to illustrate psychometric properties of the tests. SDs are close to those expected of a standardized test (i.e., 15 points)

and control groups, $t(59)=37$, $p<0.0001$, but again a medium and a small effect size, respectively, were observed for the treatment ($d=0.6$) and control conditions ($d=0.15$).

Baseline RAI scores correlated significantly with baseline mathematics ($r=0.37$, $p<0.05$) and reading scores ($r=0.42$, $p<0.05$), suggesting that relational skills are at least in some way functionally related to academic attainment. In order to further investigate the relationship between post-intervention score changes and relational ability, a series of linear regressions were conducted on data collected from the treatment group. It was found that post-intervention mathematics score changes were predicted by baseline mathematics scores ($r^2=0.15$, $p=0.002$), but not by baseline RAI score or RAI change score. Similarly, post-intervention reading score changes were predicted only by baseline reading scores ($r^2=0.23$, $p<0.001$), but not by baseline RAI score or RAI change score. Further correlational analyses revealed that baseline ability showed an inverse relationship with these score changes for both academic domains (mathematics: $r=-0.39$, $p=0.002$; reading: $r=-0.48$, $p<0.001$).

Discussion

This study found that a MET intervention in derived relational responding produced statistically and, we would suggest, educationally significant gains in mathematics and reading content areas over those achieved in the absence of an intervention. This is consistent with a growing body of work which seems to show that the enhancement of relational framing skills can have a significant and positive impact on specific measures of literacy and numeracy (Beck et al., 2023; May et al., 2022).

The current study is one of the largest conducted to date on MET training for relational skills, involving a total cohort of fifty-nine participants that were well matched and assigned at baseline (cf. Roche et al., 2023). In addition, adherence was at 100%, insofar as every student completed the training protocol, and only one participant did not complete the post-intervention assessments. In effect, the current findings cannot be attributed to factors inherent in a self-selection for participation in then research. Furthermore, allocation to groups was blind to the experimenter, as it was to the teachers that engaged in assignment of participants two conditions and the scoring of assessments was automated. Finally, and perhaps most importantly, post-intervention group differences in standardized scores were robust even when controlling for baseline scores (cf. Roche et al., 2023).

The importance of these findings is underscored by the stagnation of scholastic aptitude in the areas of reading and mathematics in the USA since the 1970s (U.S. Department of Education, 2013). In the current study, significant gains were made in both of these domains in a relatively short period of time, using a targeted intervention which focused not on curriculum content, but on the core learning skills that underlie the acquisition of such content. This is not the first study to find such effects in an educational context (see Cassidy et al., 2016; Hayes & Stewart, 2016) but it is the largest and most robust of the few studies that have employed standardized or state sanctioned indices as outcome measures.

We would suggest that, given the foregoing, the current data represent evidence of far transfer for relational skills training interventions. More specifically, there is no obvious grounds on which to argue that the current intervention “trained to the test,” because there are no items in common across the SMART intervention and the CDT assessments. The format of the tasks in the SMART training differed from those on the CDT assessments, and in fact used only nonsense words as relata. Thus, the gains in CDT scores produced must have been the result of improvements on the formally dissimilar SMART training tasks, the very definition of far transfer. However, it is worth reminding the reader that the current tested methodology is a commercially available product developed by two authors of the current study (BR and SC). Indeed, it was recently noted in a systematic review of SMART by May et al. (2023) that more studies are needed to investigate the effects of SMART independent of its creators. This is just good practice that generally protects against experimenter bias in interpreting nuanced data. Thankfully, however, there is already a large handful of independent studies that have assessed the effects of SMART in both educational and other settings and a relatively healthy ongoing series of independent studies that may inform the literature further in the near future. Nevertheless, a declaration of conflict of interest has been provided to the editor of this journal in relation to this matter.

There are a few nuanced features of the current findings that require further consideration. Firstly, while the main analysis of covariance clearly suggested a strong effect on mathematics and reading for the treatment group over the control group, post hoc exploratory analyses revealed that the control group also showed modest but significant gains in both domains over time. This outcome is most parsimoniously interpreted in terms of practice effects. While test–retest reliability coefficients for the CDT assessments have not been published, pre- and post-assessments were administered here at a considerably shorter interval than would be typical in a classroom setting under normal teaching conditions. However, it is also the case that according to the developers of the CDT assessments, these are suitable for and reliable for up to six administrations per year. Of course, it is not possible to rule out expectancy or placebo effects either, given that all participants knew they were participating in a study designed to enhance cognitive skills. This latter suggestion, however, while important to consider, does not fit well with the results of similar studies that have used control interventions also presented to participants as intended to enhance cognitive ability (e.g., Hayes & Stewart, 2016; McLoughlin et al., 2020b, 2020c). Indeed, it would be truly remarkable to observe changes in standardized scores of the magnitude observed here, and in other studies, that is entirely accountable for by expectancy effects. Nevertheless, this potential risk suggests the importance of controlling for expectancy effects in future studies.

Another interesting observation relates to the lack of correlation found between baseline RAI scores and CDT score gains, in conjunction with the finding of the regression analysis that the magnitude of RAI gain scores was also not predictive of CDT gain scores. This is somewhat surprising given that the RFT model of intellectual skills presented and tested here is based squarely on the idea that increases in generalized relational responding abilities should manifest as changes in aptitude scores on tests that assess skills that are essentially

relational in nature. It is also out of sync with other studies that have related post-treatment improvements in reading ability to gains in RAI scores (McLoughlin et al., 2020c). Our findings suggest therefore that there may be considerable noise in the current data, not least perhaps due to the relatively small sample size in statistical terms. However, it is worth noting that CDT score gains were in fact negatively correlated with baseline CDT scores. In effect, this study found evidence of diminishing returns with increased baseline CDT scores. This stands to reason insofar as gains necessarily become smaller as students approach maximum fluency in the domains of mathematics and reading. Thus, these results would appear to support the importance of considering baseline academic ability when evaluating the efficacy of relational skills training programs.

One possibility, given the lack of correlation between RAI gains and CDT gains, is that the RAI and CDT may diverge in terms of what precisely each is indexing. This may be expected, given that the RAI aims to assess purely abstracted foundational learning skills, whereas the CDT aims to assess academic attainment by calculating competencies on curricular content. To this extent, scores obtained from each test may be expected to diverge somewhat, and each may not be an equal predictor of gains on the other. Clearly, the issue of the relationship between these two measures is complex, not least because of one clear limitation of the RAI. Specifically, there is considerable variation in the correlation between RAI scores and full-scale IQ reported across various studies, with coefficients as low as 0.13 in one study (Cassidy et al., 2016) to 0.74 in another (Colbert et al., 2017). Moreover, the RAI has been found to have just about acceptable test–retest reliability ($r=0.81$; Colbert et al., 2017), although a more recent study has called its psychometric integrity into question (Cummins, 2023). In effect, the RAI may not yet be a sufficiently reliable and valid proxy of underlying intellectual ability to predict the potential effect of a relational skills intervention.

It is worth noting that the current study employed a cohort of participants in a narrow age-range, and it is not yet clear how this intervention might work with children of lower ages. Several published studies already outlined in this paper have demonstrated gains on other standardized measures with cohorts aged as young as 7 years (McLoughlin et al., 2020c). One of the studies did in fact use standardized assessments of reading that were both national (Drumcondra tests, Differential Ability Tests) and global (Wechsler Individual Achievement test; see Hayes & Stewart, 2016; see also Cassidy et al., 2016). However, ensuring adherence to a rather demanding intervention program with younger children requires great oversight and experimenter effort.

There are a host of variant, and possibly invariant factors, that might place boundaries on achievable gains in scholastic attainment resulting from relational skills training. Specifically, personality factors and levels of motivation may moderate intervention effects, but these effects have been found to be modest to non-existent (McLoughlin, et al, 2020c). On the other hand, a recent study found that attentional skills and baseline intelligence may account for differences in IQ gain across treatment and control groups (Roche et al., 2023). However, that study was methodologically compromised by unequal baseline scores across experimental

groups and so this interpretation is made with reservations and runs counter to what was observed in the current study.

Finally, it is worth noting that the RFT approach has much in common with a newly emerging cognitive perspective on intellectual development that has identified relational reasoning as a teachable skill set that characterizes a wide range of cognitive abilities (Alexander, 2019; Carpenter et al., 2005; Goldwater & Schalk, 2016; Goldwater et al., 2018; Halford et al., 2012; Kaufman et al., 2009). While each of these two fields have much to learn from the other, there have not been significant conceptual bridges built yet (although see McLoughlin et al., 2020a) and a deliverable intervention has not yet been developed by researchers in relational reasoning. On the other hand, other RFT has successfully employed a different method known as PEAK training to successfully enhance intellectual and scholastic ability (Dixon et al., 2014, 2021a, 2021b).

Conclusion

Given the current findings, it is exciting to consider what effects ongoing SMART training may have on educational progress across the longer term. This is a pressing research question which needs to be addressed using larger sample sizes, more diverse participants in terms of age and educational profile and active control groups. At present, however, it would appear that a multiple exemplar relational skills training system such as SMART may serve as a useful support to teachers in helping children to maximize their educational potential.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10864-024-09559-3>.

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Declarations

Conflict of interest This research was conducted at Wilkes University by Charles Stricker under the supervision of Jin-Joy Mao. Sarah Cassidy and Bryan Roche declare a conflict of interest involving their shareholding in a company that provides relational skills training online for commercial gain.

Informed Consent Permission from the district's superintendent and building principal was obtained by following the research procedures approved by the Wilkes University's IRB. The researcher also obtained consent from the teachers involved in the study, as well as permission from the parents or guardians of the students participating in the study, and assent from all participants.

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